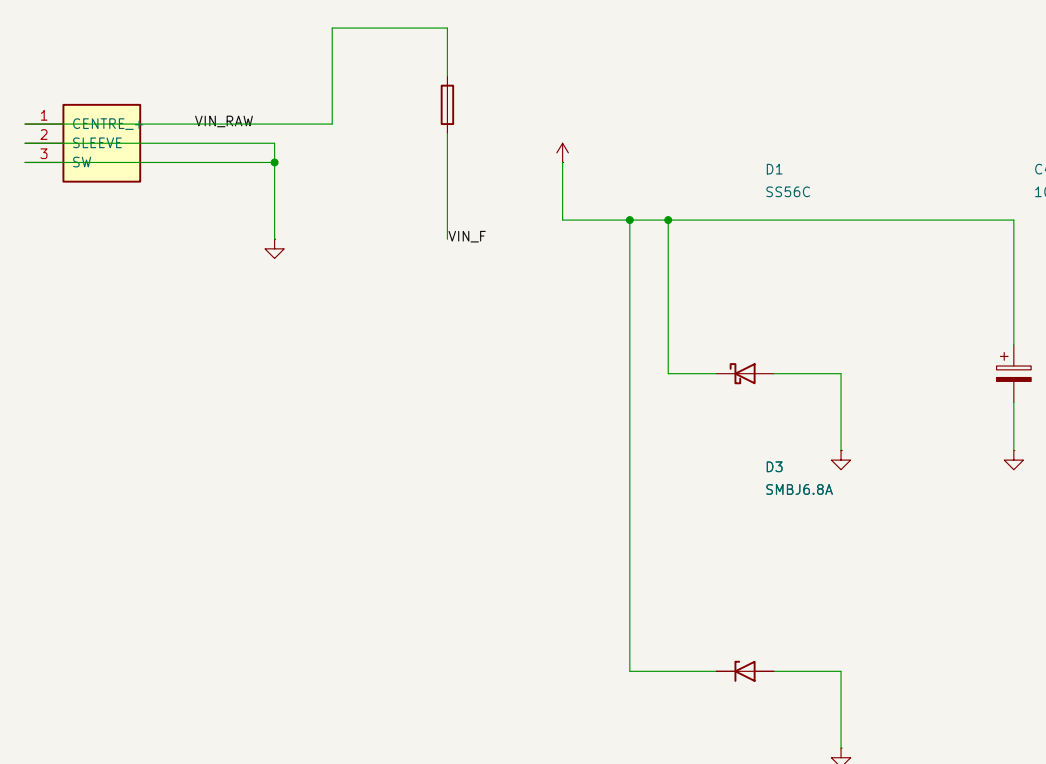
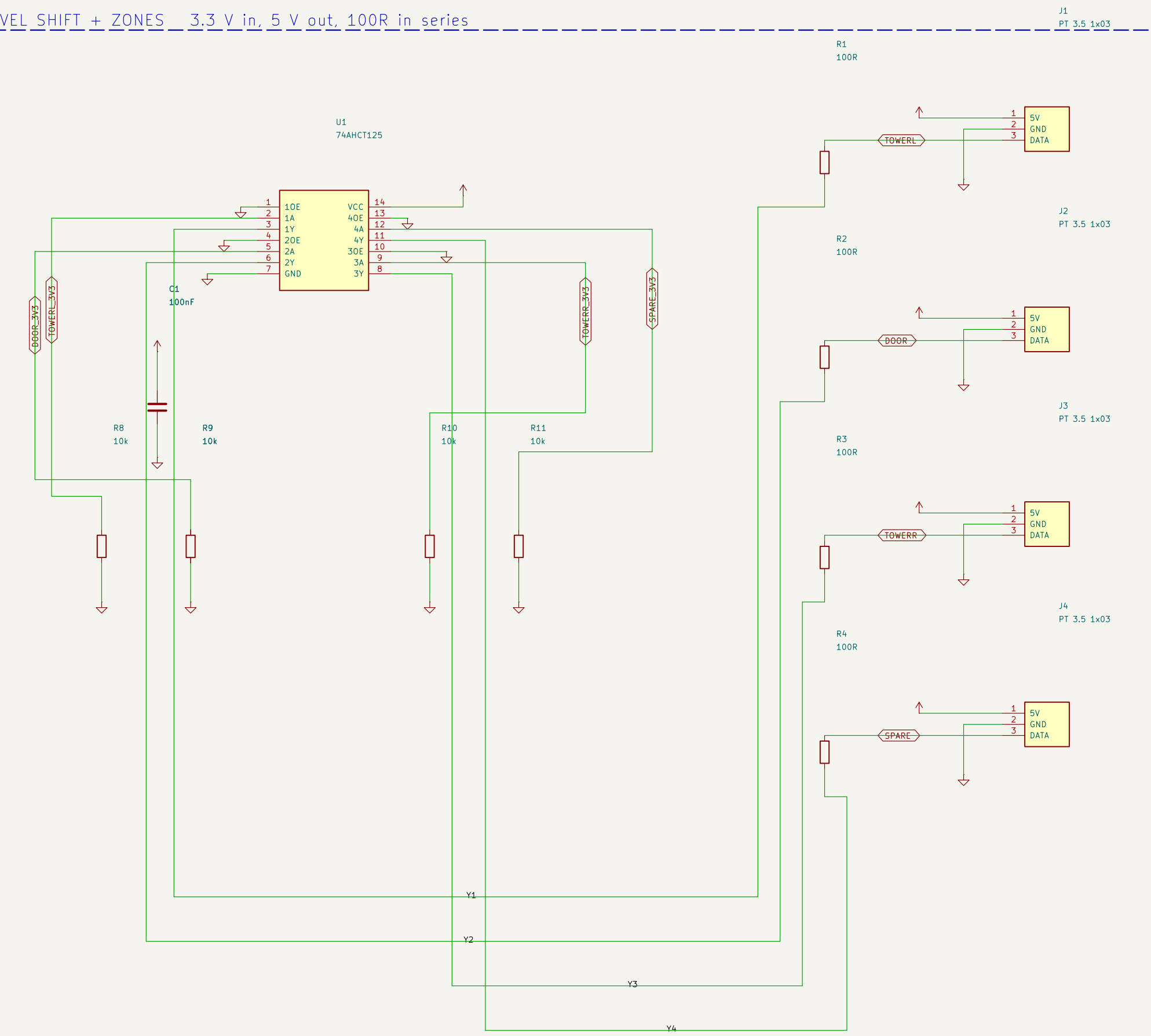
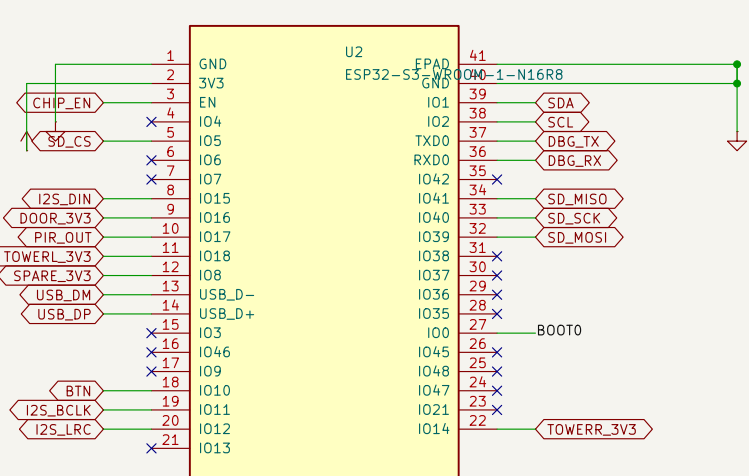


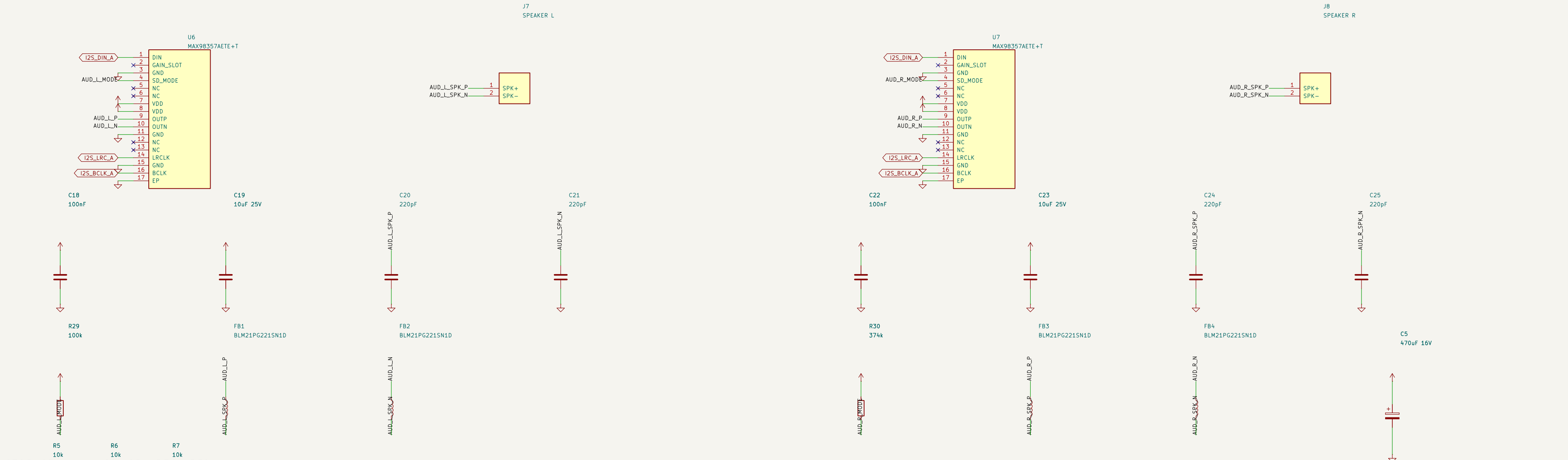
POWER /J1 -> F1 -> VIN-F: D1 clemB...D3 TVS + C4 on the protected +5V



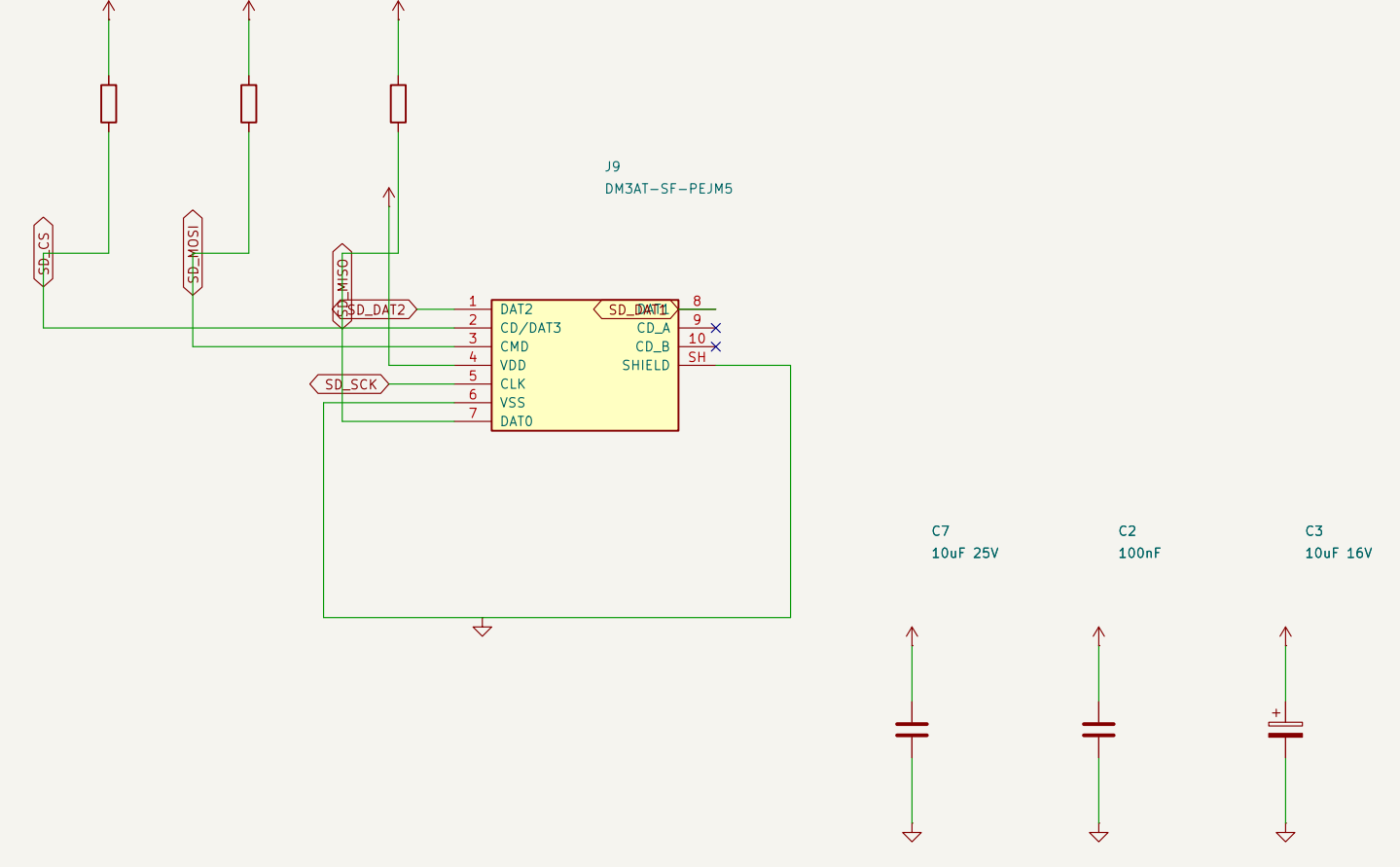
MCU: ESP32-S3-WROOM-1-N16R8: 16 MB flash / 8 MB octal PSRAMLEVEL SHIFT + ZONES 3.3 V in 5 V out_100R in series



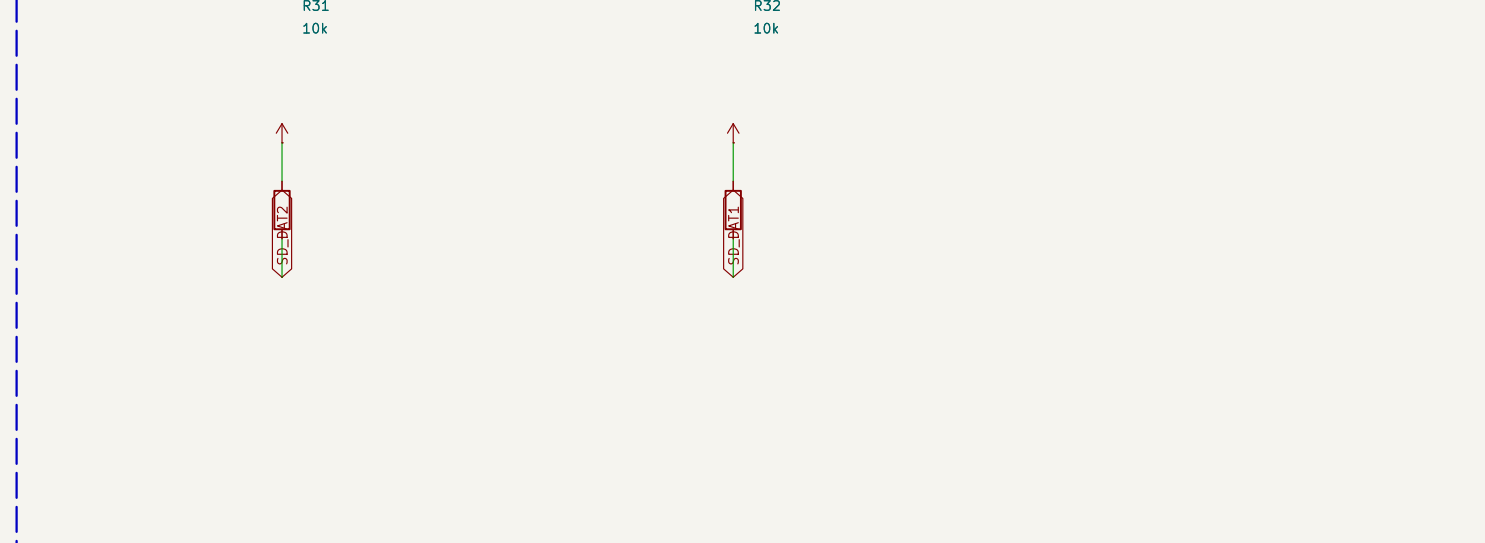
AUDIO: two on-board MAX98357A: LEFT / RIGHT stereo: 9 dB: floating speaker outputs



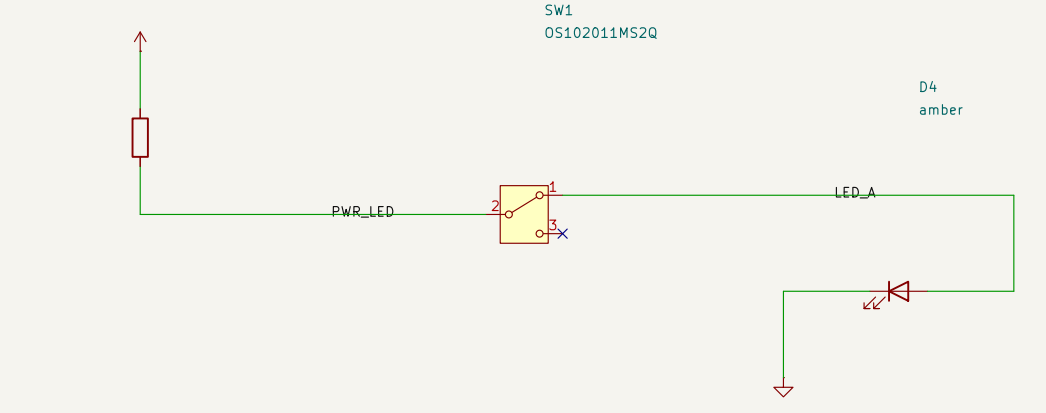
STORAGE: microSD: SPI_REQUIRED



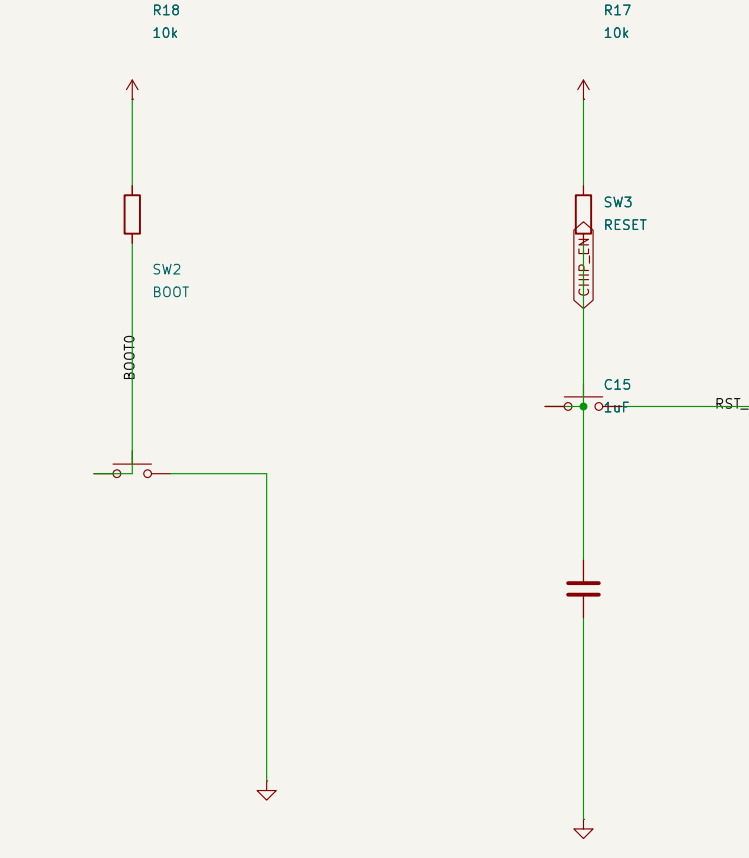
SD: unused DAT1 / DAT2 require 10k pull-ups too



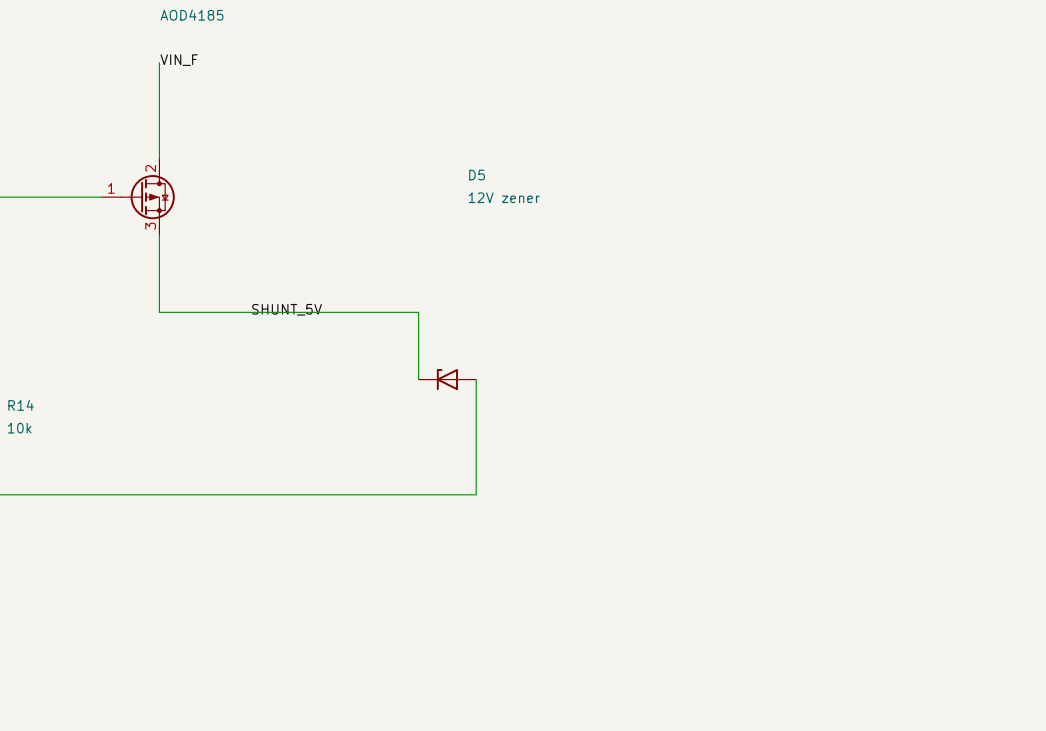
POWER INDICATOR +5V -> R12 -> SW1 -> D4



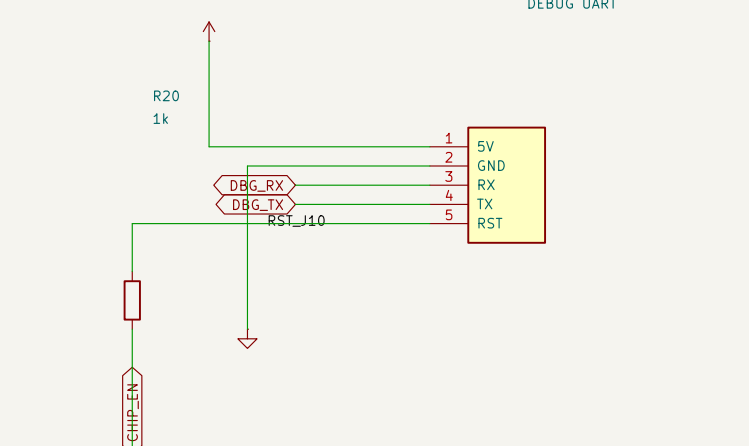
BOOT / RESET / EN: NO CAP EVER ON BOOT0



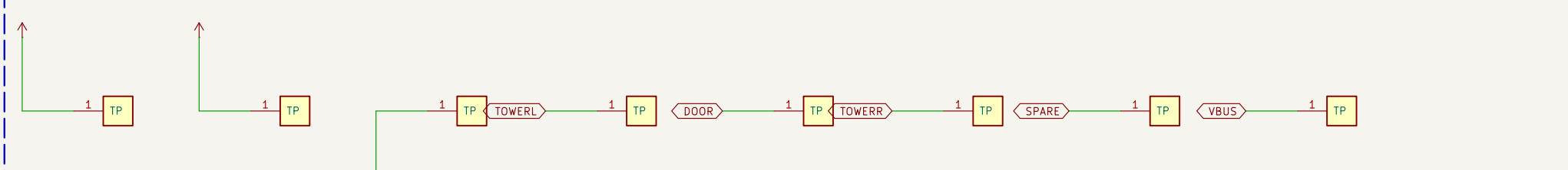
PROTECT: Q1 permanently in series: no bypass: 5 V supply ONLY



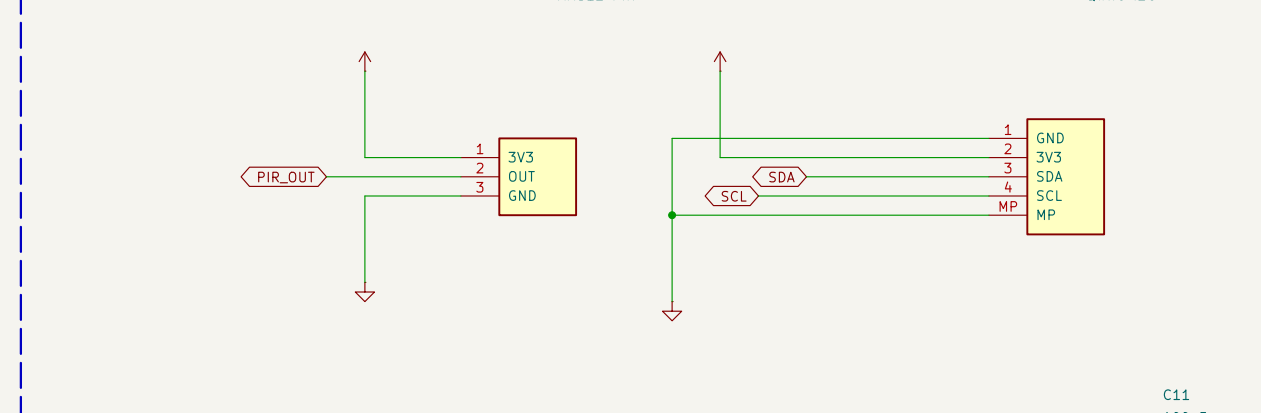
DEBUG: 3V3 UART: 5 V on PIN 1: EN via R20



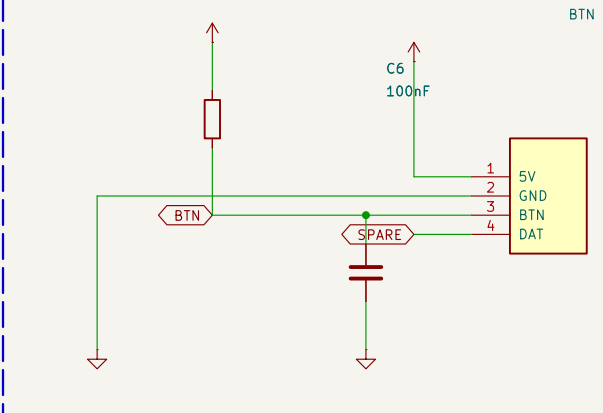
TEST POINT16 bare 1.5 mm pads: probe from the top



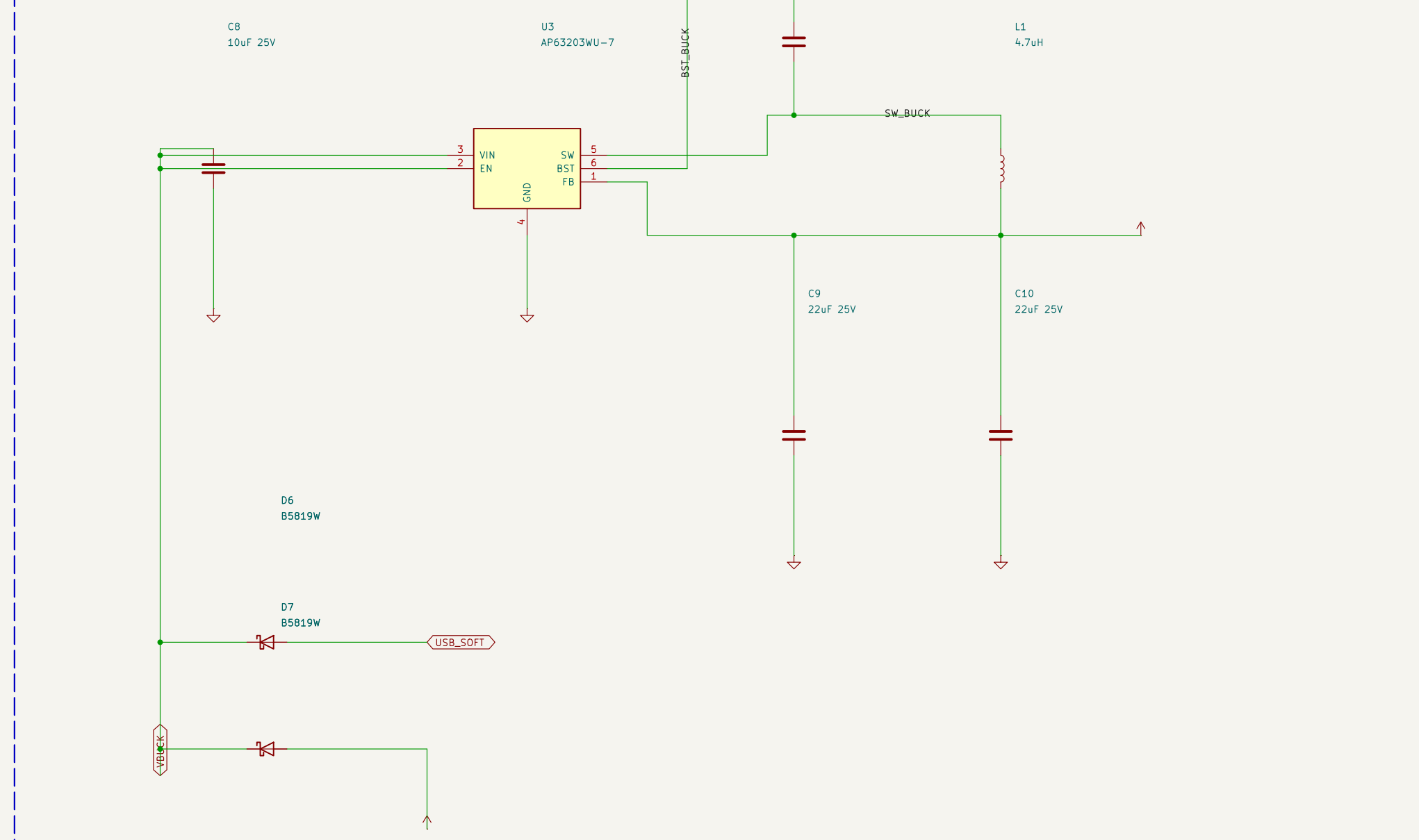
SENSOR: 1x12: 2M32 2IR and 15 to a 5V16 device: beside the on-board sensor



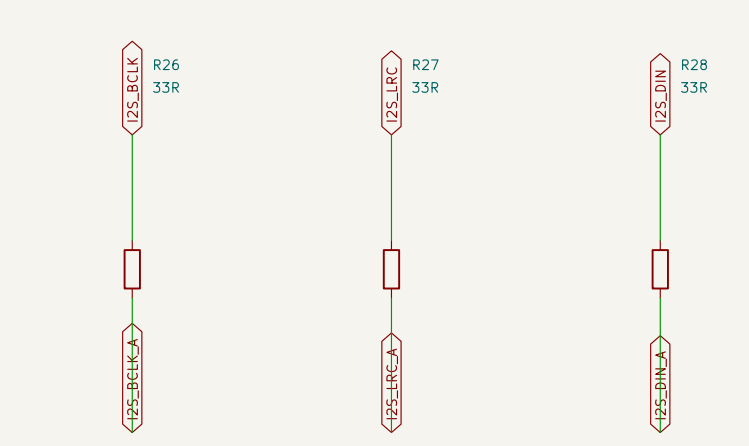
BUTTON: wires: which on D30 + one wire on SPARE



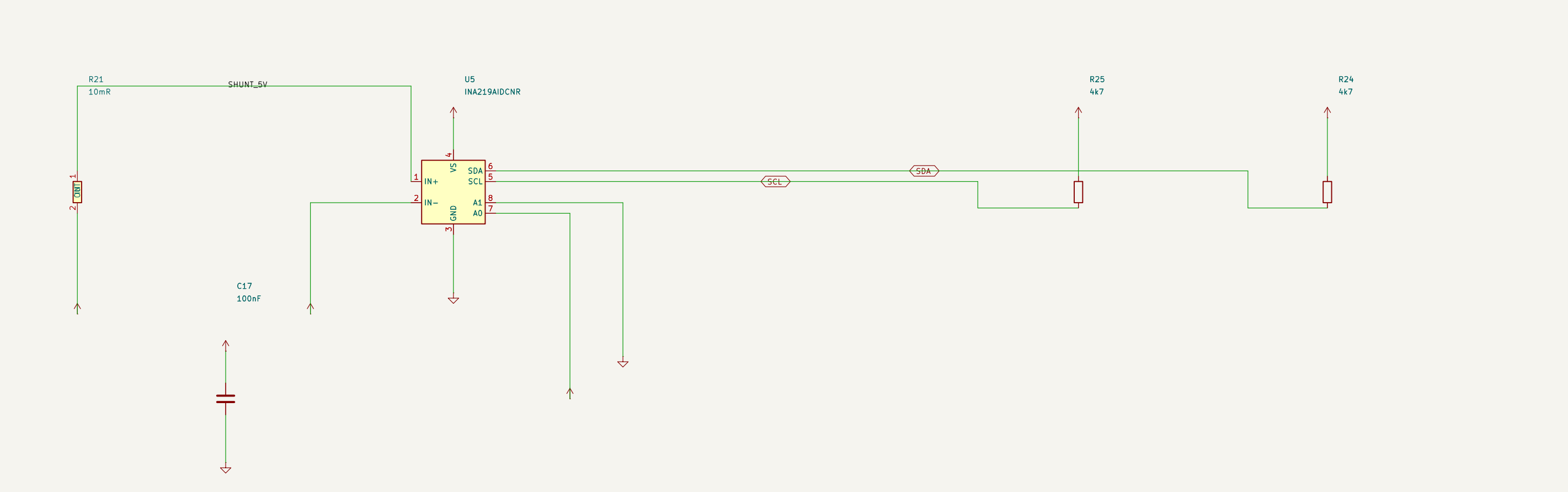
3V3: D6 / D7 OR USB VBUS and the barrel into VBUCK: U3 makes the ONE 3.3 V rail from it



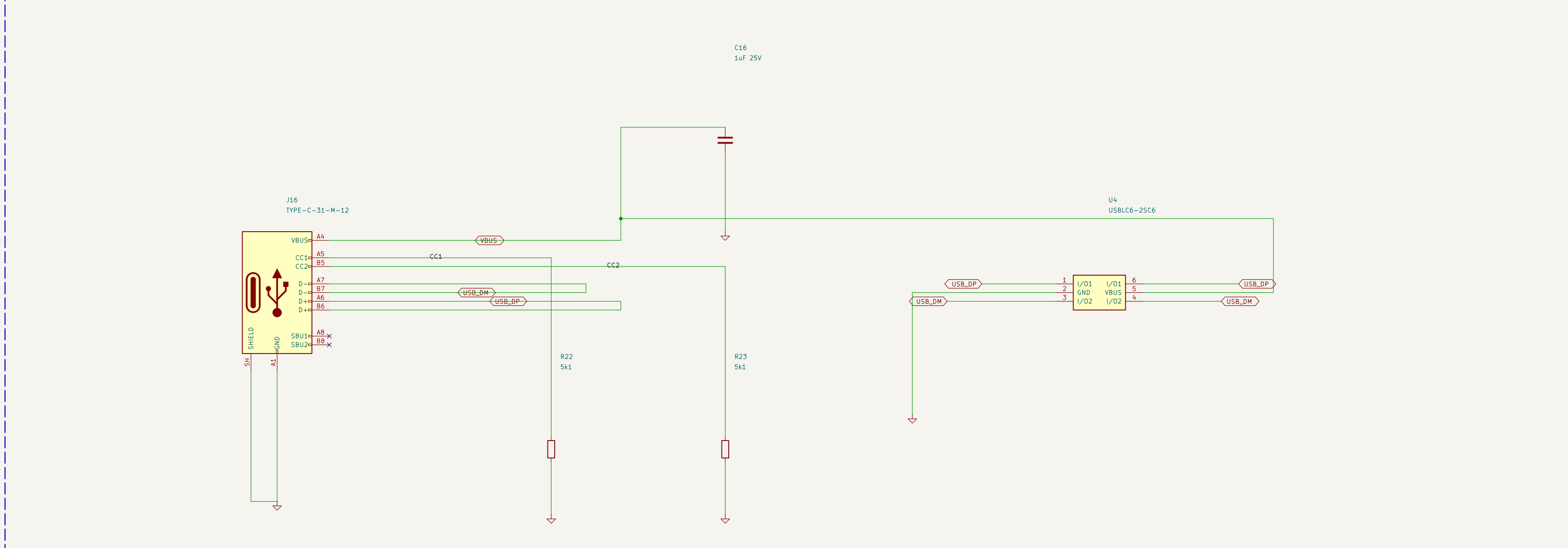
J25: 33R source damping: not a power isolation device



AMMETER: R21 10mOhm in series in the 5 V trunk: U5 across it at 0x41 and the I2C pull-ups



USB: native: no bridge chip: two CC pull-downs: a USBLC6-25C6 the pair passes THROUGH 1uF VBUS



USB STARTUP: controlled rise: QOD disabled: D6 blocks barrel backfeed

